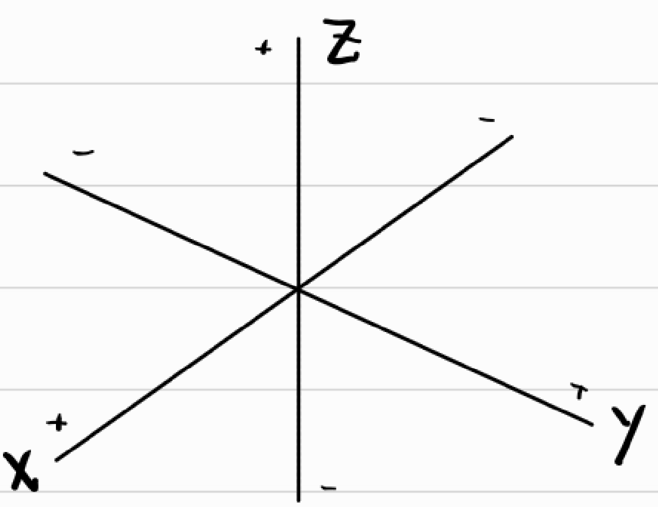
Usually In Cartesian Coordinates we denote Points as an (x, y) Pair for distances along the x and y axes.

For Three Dimensional Space the Convention is much the Same with the addition of Z - a new axis. So our coordinates are (x, y, z)

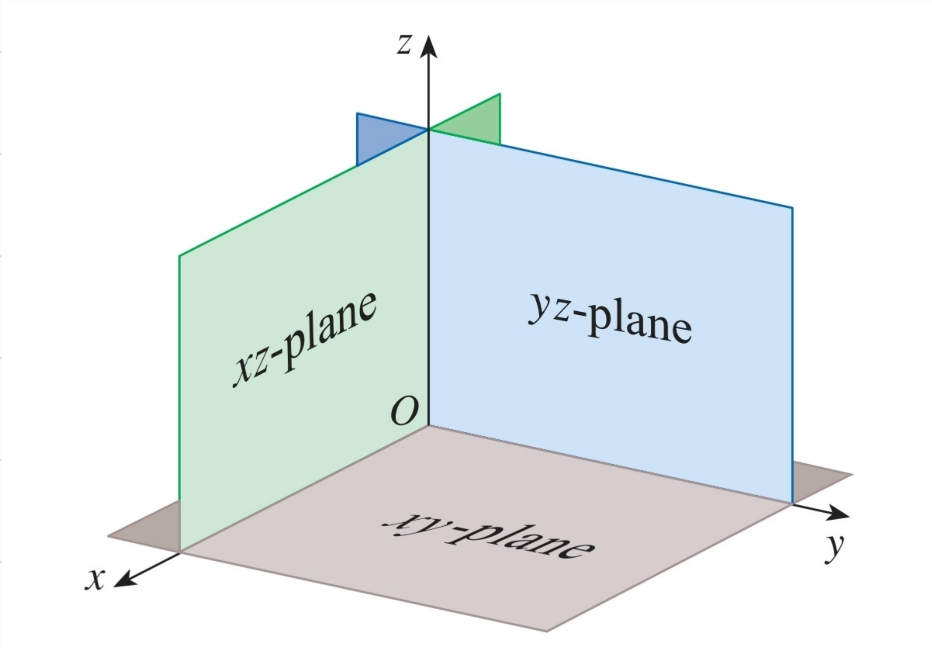
By Convention, we treat Z as the vertical, height, axis, while x and y are the base axes.

The entire Space could be turned, but the relative Position of the axes must remain the Same. The usual Presentation is below



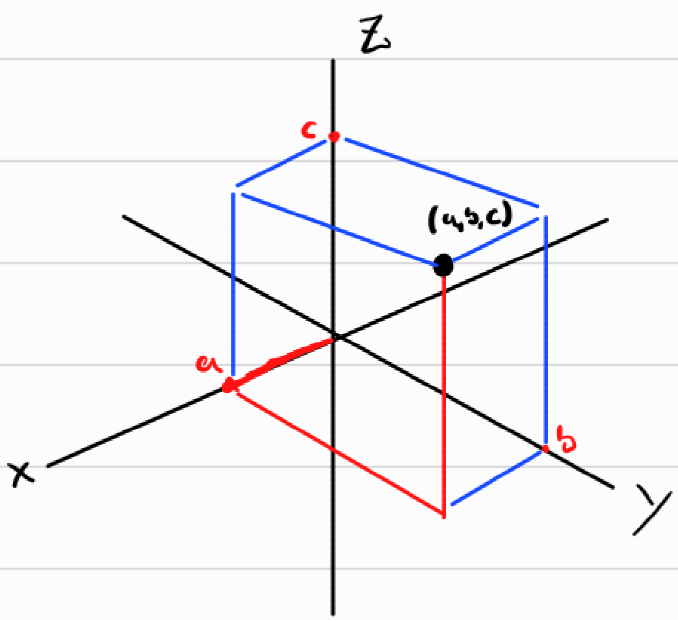
Take note of the Positive and negative axes direction.

By just considering the directions, we create the three coordinate planes



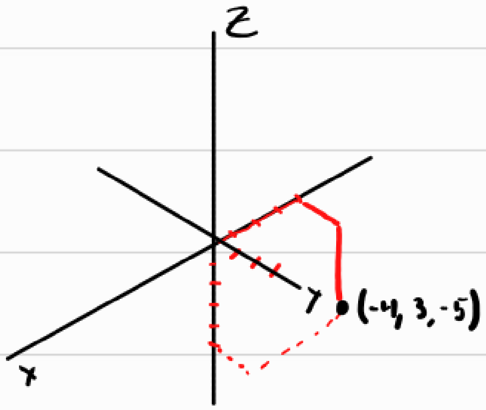
Plotting points is similar, just more difficult to visualize.

Say, a,b,c are positive, the point (a,b,c) is

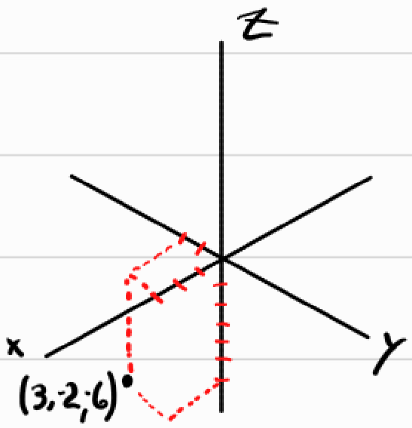


The box is to help visualize, just sketch the red lines in practice

$(-4, 3, -5)$



$(3, -2, -6)$

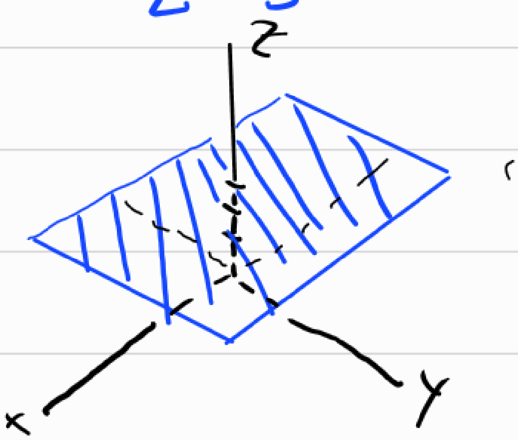


In 2D, an equation of x and y will create a curve in $\mathbb{R}^2$

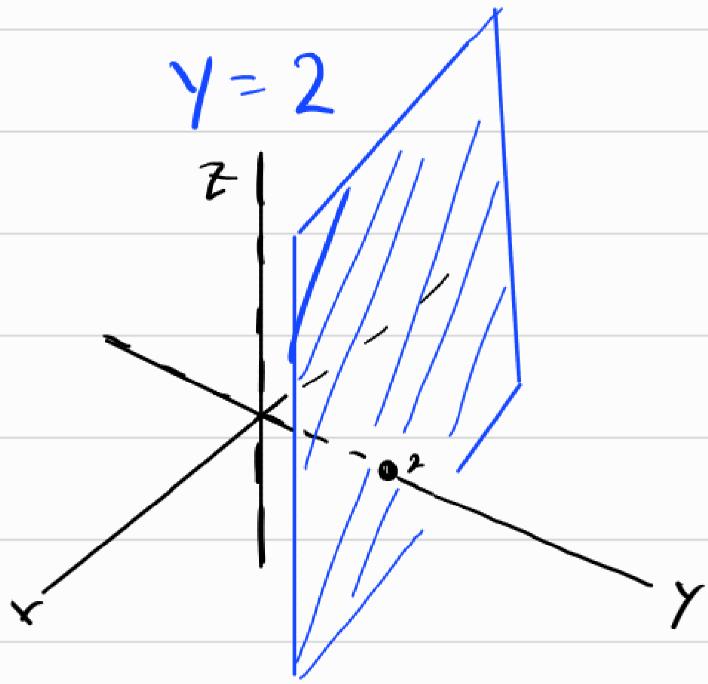
In 3D, an equation of x, y, and z will create a surface in $\mathbb{R}^3$

Ex:

$z = 3$



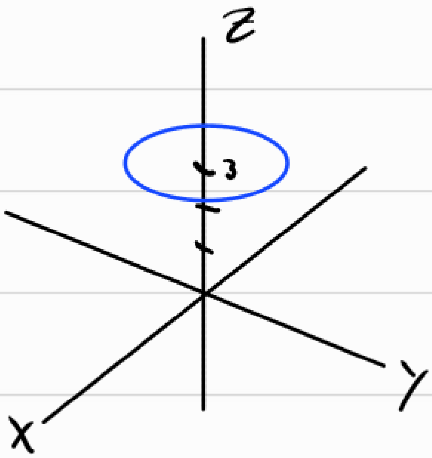
$y = 2$



Ex:

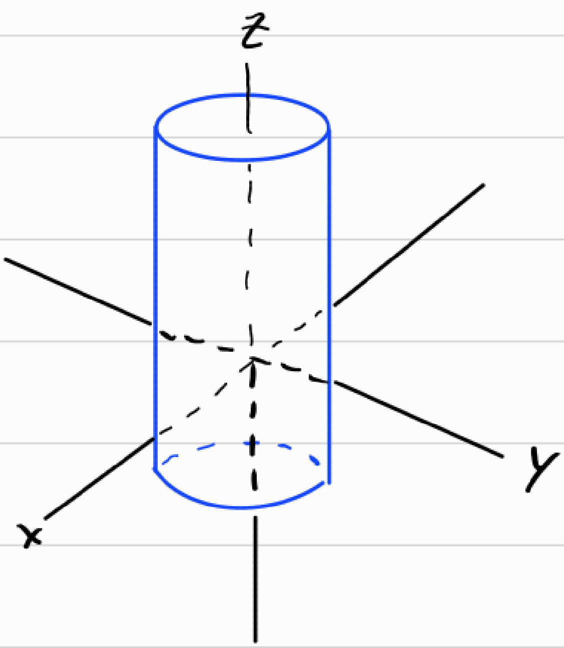
1) $x^2+y^2=4$, $Z=3$

Circle in x-y Plane



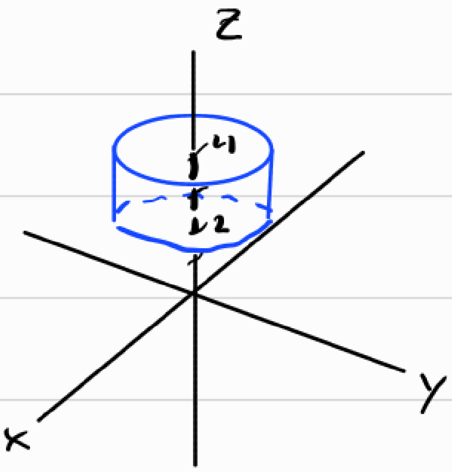
2) $x^2+y^2=4$

Here, there is no restriction on Z, so all Z are Possible



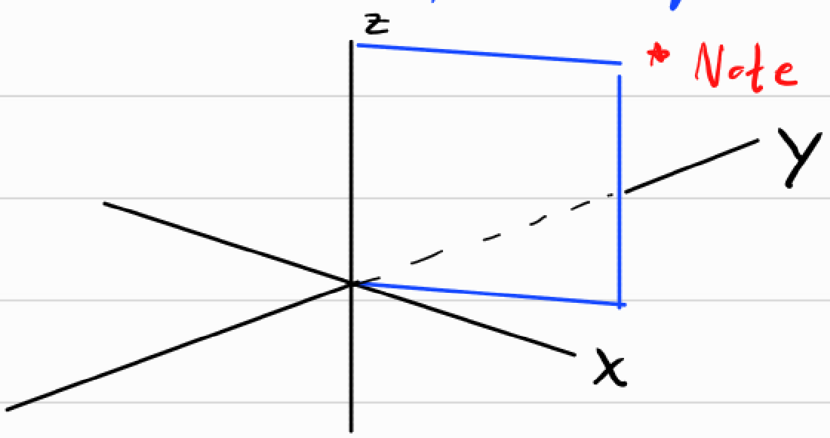
3) $x^2+y^2=4, \quad 2 \leq z \leq 4$

Similar to infinite Cylinder



4) The Plane $y=x$ for $x,z \geq 0$

* Note axis are turned



Distance

Take Points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$

The distance between those points is

$$|P_1P_2| = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

From this formula, pick some point (h, k, l) and radius r,

the points (x, y, z) such that

$$\sqrt{(x-h)^2 + (y-k)^2 + (z-l)^2} = r$$

form a sphere, its not hard to see this as

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

Ex: A sphere centered at $(3,-1,6)$ that passes through $(5,2,3)$.

$$r = \sqrt{(3-5)^2 + (-1-2)^2 + (6-3)^2}$$
$$= \sqrt{4+9+9}$$
$$= \sqrt{22}$$
$$r^2 = 22$$

Sphere

$$(x-3)^2 + (y+1)^2 + (z-6)^2 = 22$$

Ex:

$$x^2 + y^2 + z^2 + 4x - 6y + 2z = -6$$

$$x^2 + 4x + y^2 - 6y + z^2 + 2z = -6$$

$$x^2 + 4x + 4 - 4 + y^2 - 6y + 9 - 9 + z^2 + 2z + 1 - 1 = -6$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = -6 + 4 + 9 + 1$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = 8$$

Sphere w/ center $(-2,3,-1)$ and radius 8